Angel Patricio

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Education

Massachusetts Institute of Technology — **Cambridge, MA**
M.Eng. (admitted), Electrical Engineering and Computer Science — *Entry Deferred*

Massachusetts Institute of Technology — **Cambridge, MA**
B.S., Artificial Intelligence and Decision Making (EECS, Course 6-4) — *2022–2026*
Grad. Coursework: Deep Learning, Reinforcement Learning, Statistical Learning Theory, Nonlinear Optimization. **Also:** NLP, Real Analysis

Publications

J. Yuan, B. Le, K. Le, Y. Shi, A. Singh, R. Sharma, **A. Patricio**, R. Raskar. SONAR Web: A Platform-Agnostic Framework for Real-Time Decentralized Learning Across Heterogeneous Edge Clients. *ACM MobiCom FLEdge-AI Workshop*, 2025. **Best Presentation Award.**

Independent Research & Projects

The Mechanistic Function of the Recurrent Block in Looped Transformers — **[code] [blog]**
Independent Research · PyTorch, einops, cloud GPUs (Vast.ai) — *Oct 2026–*
- Examined computation in one-layer, plain looped, and Huginn-style transformers on n-hop addition mod 113 as a precursor to downstream studies on how architectural decisions impact monitorability.
- Found that a linear probe on the plain loops' hidden state reaches 90% accuracy on the answer 2–6 passes before the model's output does.
- Found every model averages its inputs, then reads out the answer. However, the loops differ in behavior: plain loops' performance collapses past training depth (~100% @ 8 $\rightarrow$ ~1% by pass 256); one Huginn-style model examined even learned to ignore its carried state.
- Wrote up the results on my blog; released code and results for all 44 runs, with every figure reproducible from a fresh clone.

IntAct: Interpretable Activations for Reward Hacking — **[code] [blog]**
Independent Project · vLLM, PyTorch, transformers, PEFT — *Sep 2026–*
- Designed and shipped an interpretability tool for reward hacking. Samples and grades rollouts from RL-trained Qwen3-4B adapters (rl-rewardhacking: ariaw, Engels, Nanda), extracts activations around the token where hacking begins, and loads them via a Python API.
- Closely replicated upstream's hack rate (73.7% vs. ~79% reported); activations match generation-time values within 4e-5 relative error.

The Edge of Stability for LLM Post-Training — **[code] [blog]**
Course Project, MIT 6.7960 Deep Learning · PyTorch, transformers, PEFT — *Fall 2025, Revisited 2026*
- Investigated a null result from my course project, tracing it to four measurement flaws. Directed a redesign with pre-registered tests.
- Found fp32 weights lost a median 3.4% of each update to rounding, so moved to fp64; cross-checked 2 custom eigensolvers within 1%.
- A small MLP control failed the $\lambda_{\max}$ test, but step curvature held near $2/\eta$; scoped new tests and reported Pythia-70m as inconclusive.

Undergraduate Research

A Benchmark for Spatial Reasoning with Origami — **Cambridge, MA**
MIT Computer Science and AI Lab (CSAIL), Computer Assisted Programming Group (Solar-Lezama) — *Feb 2025–Aug 2025*
- Surveyed existing evaluations for spatial reasoning in neuro-symbolic and language models, presenting findings and ideas at weekly group meetings; acknowledged in OrigamiBench (ICML LM4Plan '26).
- Built an early prototype: extended an inverse-origami MDP environment to 3D `.obj` targets and ran its baselines on ~25 of them.

SONAR Web: Real-Time Decentralized Learning Across Heterogeneous Edge Clients — **Cambridge, MA**
MIT Media Lab, Camera Culture Group — *Nov 2024–Feb 2025*
- Wrote the first Python (PyTorch) and browser (TensorFlow.js) clients for a decentralized-learning platform, later extended and published.
- Proposed a JSON weight format, chunked weights past WebRTC's 16MB limit, and trained end to end on 4 topologies (up to 12 peers).

Masked Language Modeling for Genomics — **Cambridge, MA**
MIT Institute for Medical Engineering and Science (IMES), Edelman Lab — *Apr 2024–Aug 2024*
- Scaled the lab's DistilBERT masked-LM pipeline for single-cell expression from Tabula Sapiens to the 30M+-sample CELLxGENE census.
- Moved training from one A100 to 8× RTX 4090s with DDP, cutting epoch time from over an hour to ~15 min; added WandB sweeps.

Technical Skills

ML: Transformers, LLMs, deep RL, LoRA/PEFT, multi-GPU (DDP)
Frameworks: PyTorch, JAX, transformers, vLLM, TensorFlow.js
Languages: Python, JavaScript, Julia, LaTeX
Interpretability: Probing, activation patching, steering
Libraries: NumPy, pandas, einops, scikit-learn, TransformerLens, verl
Tools: Git, Linux, WandB, Vast.ai, pytest, Claude Code